

Introduction to computing

Operating systems:-

- Operating systems are the souls of computer.
- They decide the behaviour of computers and the interactions you have with them.
- For instance, if you are a windows user normal mode of interaction is a GUI.
- For linux users, the shell is much more common.

What do OS do?

API

System calls and application programming interface.

API

Application → to another application

Application → OS

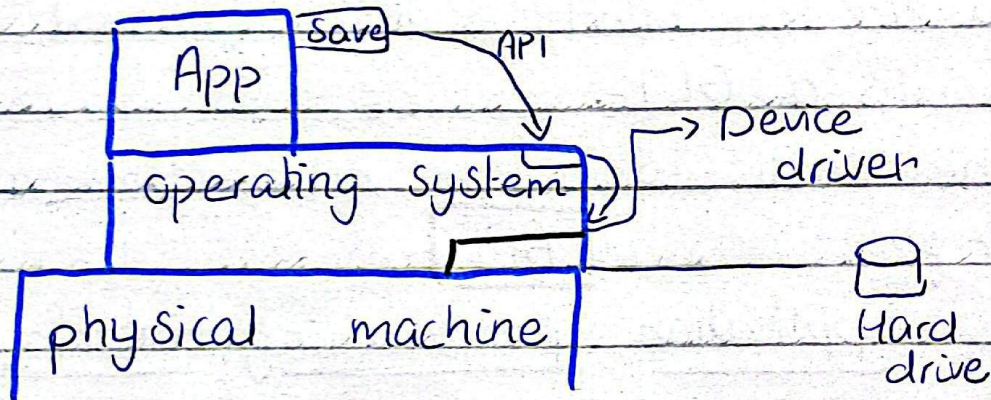
user systemcall and

API

↓
How ever our application
interact with OS.

The role of an OS:-

- MS-excel save button. e.g:-



The social computer:-

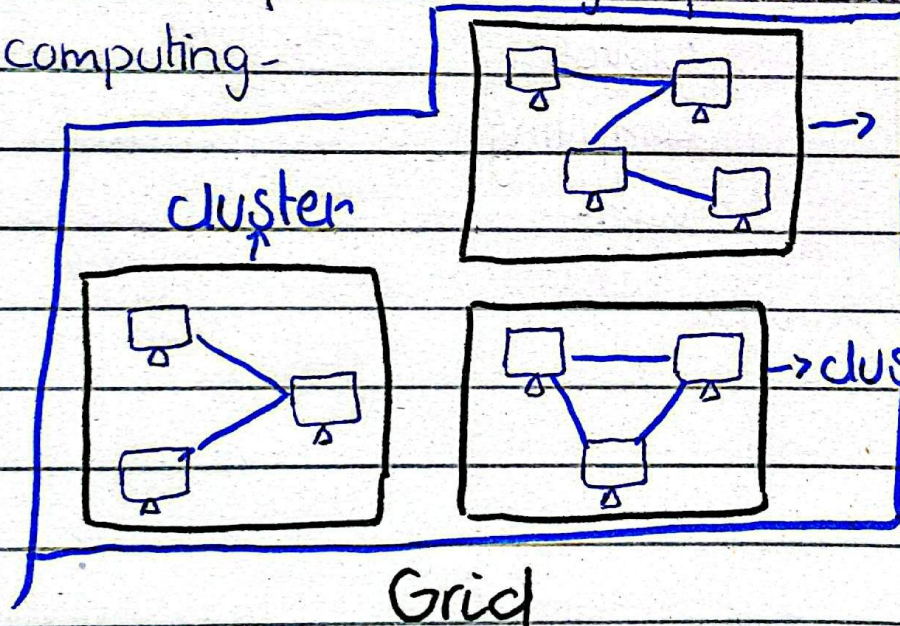
- Society of computer
- Computers do not operate in isolation. e.g:-
 - Some comp good in storage
 - Some good in power.
- Some computers are good at one function, others at some different one.

Cloud computing :-

→ You might have heard about super computers.

↳ for bigger work.

• The immense power of super computers can be emulated with the concepts of High performance computing -



• Cluster computing and grid computing puts several comps together to create powerful functions -

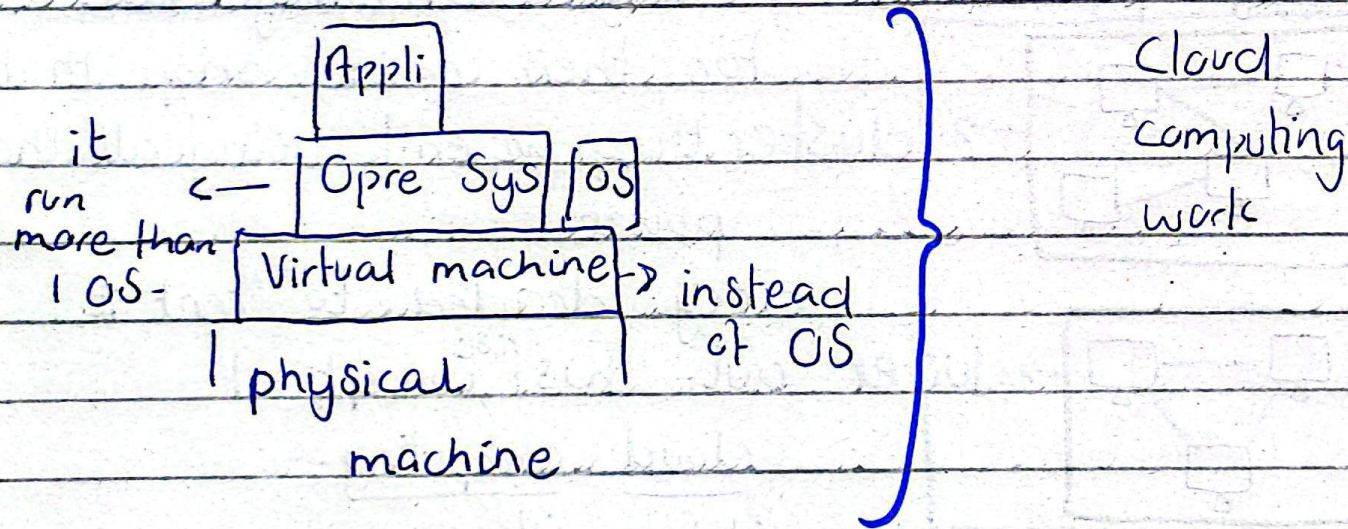
• Amazon created such a large computing base for their online shop that they weren't using all the power.

• They decided to rent it out thus ^{rise} concept of cloud computing.

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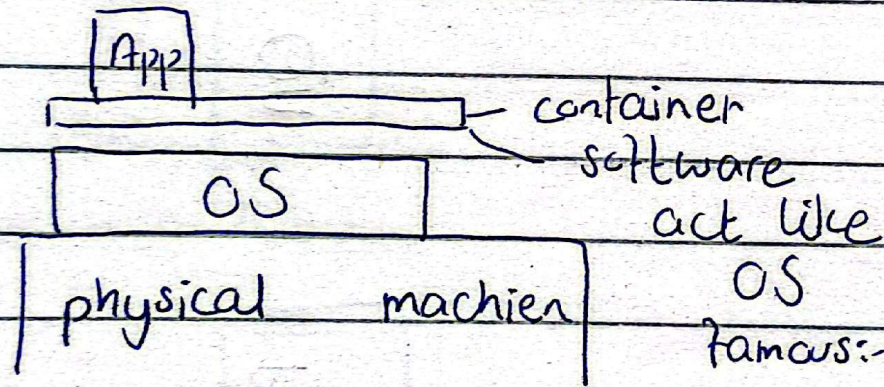
Truth behind cloud computing:-

- Recall what a 'machine' does
- What's stopping you from writing a machine that acts like the hardware you have.
- This is called a virtual machine.

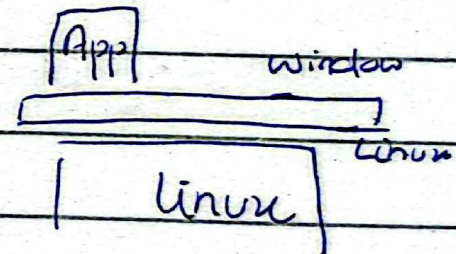


VMs versus containers:-

- Containers (Docker) are light weight virtual machines (sort of)
- Create once, deploy everywhere



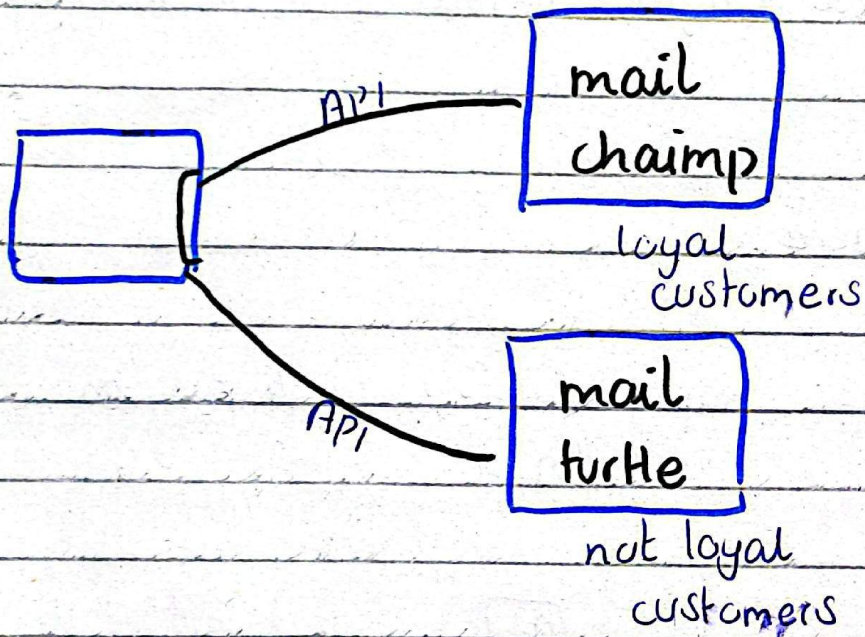
Window appli run on linux



Docker

Network APIs :-

Typical workflow in
production environment:-



Developer
sandbox
you pass
all tests
there

QA

Production
department